

EcoTrace – Complete Frontend & UI/UX Design Specification (v1.0)

Design Vision

Create an emotionally engaging, premium, futuristic sustainability platform that makes carbon tracking feel like a game rather than a calculator.

Keywords:

- Futuristic
 - Minimal
 - Premium
 - Climate-focused
 - Gamified
 - Emotional
 - Dark Neon Aesthetic
-

Design Language

Theme

Dark Nature × Cyber Neon

Inspiration:

- Apple Health
 - Notion
 - Stripe Dashboard
 - Duolingo Gamification
 - Arc Browser Animations
 - Spotify Glassmorphism
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Color System

Background

Forest Night: #0A0F0D Dark Moss: #111A14 Elevated Card: #162019

Primary Accent

Neon Lime: #39FF14

Secondary Accent

Mint Teal: #00E5A0

Warning

Amber Sun: #FFB830

Danger

Carbon Red: #FF4D4D

Text

Primary: #E8F5E2 Secondary: #6B8F71 Border: rgba(255,255,255,0.08)

Typography

Display Font: Space Grotesk Bold

Body: Inter Regular Inter Medium

Data: JetBrains Mono

Font Scale:

Hero Heading: 64px

Section Heading: 40px

Card Heading: 24px

Body: 16px

Small: 14px

Data: 18px

Global UI System

Border Radius

Cards: 24px

Buttons: 18px

Input: 16px

Orb: 50%

Shadows

Card Shadow: 0px 20px 40px rgba(0,0,0,0.35)

Neon Glow: 0px 0px 30px rgba(57,255,20,0.4)

Danger Glow: 0px 0px 30px rgba(255,77,77,0.3)

Navigation

Height: 80px

Layout: Logo Left Links Center CTA Right

Menu Items: Home Dashboard Tracker Insights Leaderboard

CTA: Start Tracking

Mobile: Hamburger Drawer

Sticky Navigation: Enabled

Glass Effect: backdrop-filter: blur(20px)

Landing Page (Hero)

Height: 100vh

Layout: 2 Columns

Left: Headline Subheading Buttons

Right: 3D Animated Carbon Orb

Headline:

Track Your Carbon. Change Your Future.

Subheading:

Understand your environmental impact and build greener habits one day at a time.

Buttons: Primary: Start Free

Secondary: Watch Demo

Background: Dark gradient Floating particles Subtle animated leaves

Hero Orb Design

Size: 420px desktop 300px tablet 240px mobile

States:

Green: <2 tons

Amber: 2–5 tons

Red:

5 tons

Animations:

Pulse: 4s infinite

Breathing Scale: 1 → 1.05

Particle Orbit: 20 floating particles

Mouse Parallax: Enabled

Shadow Intensity: Depends on score

Onboarding Quiz

Layout: Multi-step Wizard

Progress Bar: Top

Steps:

1. Transport
2. Diet
3. Home Energy
4. Shopping & Flights

Each Step Contains:

Illustration Question Multiple Choice Cards

Card Size: 320x150

Card States: Default Hover Selected

Buttons: Previous Next

Transition: Slide animation 400ms

Carbon Reveal Screen

Full Screen Modal

Sequence:

Loading Animation Calculating... Orb Expands Score Appears

Display:

Your Annual Footprint

3.2 Tons CO₂

Comparison Badge:

23% Lower Than Global Average

Buttons: Go to Dashboard

Celebration: Particle Explosion

Dashboard

Desktop Layout:

2 Columns

Left: Carbon Orb Statistics

Right: Charts

Tablet: Stacked Layout

Mobile: Single Column

Dashboard Widgets

Carbon Score Card

Height: 320px

Contains: Orb CO₂ Value Status Last Updated

Category Breakdown Card

Chart: Donut Chart

Categories:

Transport Diet Home Travel Shopping

Center Text: Total CO₂

Weekly Trends

Chart: Smooth Line Chart

X Axis: Days

Y Axis: CO₂

Hover Tooltips: Enabled

Global Comparison Card

Design: Horizontal Progress Card

Shows:

You: 3200kg

World: 4700kg

Percentage Difference

Daily Action Tracker

Layout: 2 Columns

Left: Green Actions

Right: Carbon Actions

Cards: 160x140

Contents:

Icon Action Name CO₂ Impact Toggle Button

Interactions:

Click: Add Action

Orb Updates: Real Time

Microanimation: Bounce

Streak Widget

Circular Counter

Displays:

 11 Day Streak

Progress Ring: Animated SVG

Glow: Neon Lime

Insights Section

Desktop: 3 Columns

Mobile: 1 Column

Cards:

Icon Insight Title Description CO₂ Savings Try This Button

Examples:

Switch to Public Transport Save 500kg/year

Reduce Meat Consumption Save 700kg/year

Air Dry Laundry Save 100kg/year

Button: Add to My Plan

Personal Action Plan

Kanban Style

Columns:

To Try In Progress Completed

Drag and Drop: Enabled

Completion: Confetti Animation

Goal & Offset Calculator

Cards:

Monthly Goal Trees Required Solar Offset

Calculations:

1 Tree $\approx$ 22kg CO₂/year

Progress Bar: Animated

Goal Achieved: Confetti Explosion

Community Leaderboard

Table Layout

Columns:

Rank Name CO₂ Saved Badge

Current User: Highlighted

Top 3: Podium Cards

Gold Silver Bronze

Share Button: Instagram Story Card

Badges System

Cards:

🌱 First Log ⚡ 7 Day Streak 🌍 1 Ton Saved ✈️ Flight-Free Month 🥗 Vegan Week

Animation: Badge Flip

Footer

Sections:

Product Features Privacy GitHub Contact

Quote:

Every gram counts. Start now.

Mobile Design

<480px

Single Column

Bottom Navigation:

Home Dashboard Track Insights Profile

Floating Action Button:

- Log Action

Orb: 240px

Cards: Full Width

Charts: Swipeable

Motion Design

Duration: 250ms

Easing: ease-out

Animations:

Fade Up Scale In Slide Left Orb Pulse Particle Float Glow Transition Card Lift

60fps Required

Glassmorphism

Cards:

Background: `rgba(22,32,25,0.75)`

Blur: 20px

Border: 1px `rgba(255,255,255,0.08)`

Accessibility

Contrast Ratio: WCAG AA

Keyboard Navigation: Supported

Focus States: Visible

Screen Reader Labels: Added

Reduced Motion: Supported

Suggested Folder Structure

`src/`

`assets/ images/ icons/`

`components/ Navbar Hero CarbonOrb Quiz Dashboard Tracker Insights GoalCalculator Leaderboard Footer`

`styles/ variables.css global.css animations.css responsive.css`

`pages/ home dashboard`

UI Deliverables

1. Landing Page
2. Quiz Screens
3. Carbon Reveal Screen
4. Dashboard
5. Tracker
6. Insights
7. Goal Calculator
8. Leaderboard
9. Mobile Version
10. Design System Library
11. Component Library
12. Prototype Animations

Target Experience: A premium climate-tech product that feels like Apple Health + Duolingo + Spotify for sustainability.